

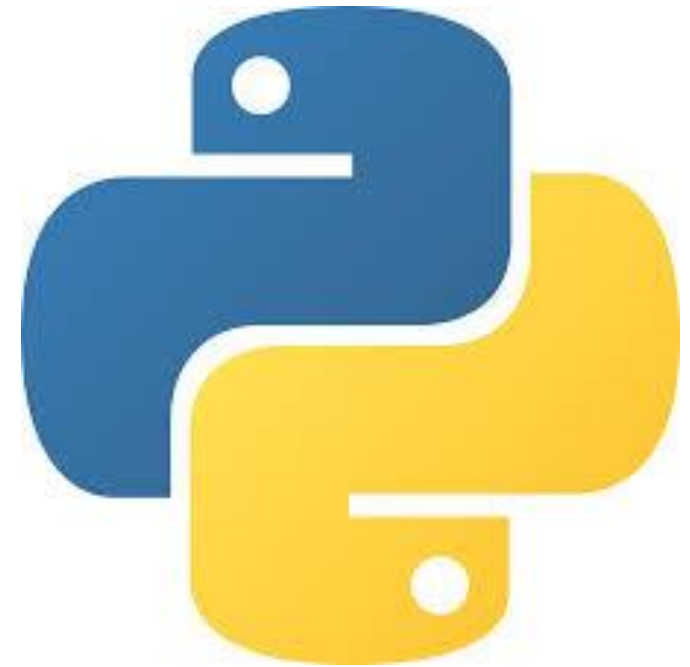
Automating Excel Spreadsheets with Python

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What is Python?

- Python is a free and open-source programming language.
- It is one of the most widely used language
 - [TIOBE Index](#)
 - [Stack Overflow](#)
- It is easy to read and write compared to other programming languages.
- Extensive Library of pre-built modules you can import into any project
 - Data Analysis
 - Encryption
 - Web Applications
 - And many more!



Getting Started

- First you will need to download an IDE and a version of Python
 - Popular IDEs:
 - [Visual Studio Code](#) – Highly customizable with multiple free extensions
 - [PyCharm](#) – Deeply integrated tools for large codebases and team projects
 - [Python IDLE](#) – Zero setup required; included with the Python installation for immediate coding

Python Libraries for Data Analysis and Automation

- There are many Python libraries available for created/edited spreadsheets:
 - Pandas – a fast and powerful data analysis and manipulation tool
 - Openpyxl – designed for reading/writing to Excel files and has access to almost all tools available within Excel.
 - Csv – designed for reading and writing csv files
 - Gspread – design for reading/writing google sheets
 - Email and SMTPlib – libraries for sending and receiving emails
 - Streamlit – displaying interactive data onto web applications
- Helpful links:
 - Python for Beginners: <https://www.python.org/about/gettingstarted/>
 - Python Tutorials: <https://www.w3schools.com/python/>
 - Automate the boring stuff: <https://automatetheboringstuff.com/>

Project 1 – Automating Account Tracking and Creating Password Alerts

- Problem 1: Users were letting their USAS/USPS passwords expire.
- Problem 2: Auditors wanted a list of all User accounts, roles, and MFA status for all USAS/USPS applications.
- Solution: Generated daily USAS/USPS user-listing reports across all applications and sent them via FTP to a server. A Python script then processed each file to identify upcoming password expirations and wrote the user information to a master user listing file.
 - The Pandas library is used to isolate each user and their password expiration date.
 - The Email and SMTP library sends emails to users whose passwords expire in 7 days.
 - Openpyxl is used to create one master file of all users.

```
users_df = pd.read_csv(input_file_path) #creates a dataframe from the input csv file containing user data
usernames_list = users_df['Username'].tolist() #creates a list of usernames from the dataframe.
```

Figure 1: Using Pandas to isolate the usernames out of a csv file

Project 2 – Automating the Creation of Excel Graphs and Data Clean Up

- Problem 1: Districts wanted visual representations of their financial data to present to their board.
- Problem 2: Some generated reports required significant cleanup before they could be used effectively.

A	B	C	D	E	F	G	H
Full Account	Description	Active	MTD Received	MTD Expended	Month End Balance	Month Name	Fiscal Year
001-0000	GENERAL FUND	TRUE	[2054442.02, 760288.38, 800024.10, 83	[958669.77, 1048180.86, 944780.01, 88	[8220181.05, 7932288.57, 7787532.66,	[July, August, September, October, No	2022
001-0000	GENERAL FUND	TRUE	[1995530.48, 776006.48, 854310.89, 11	[974336.55, 1051875.93, 929589.32, 91	[8610641.39, 8334771.94, 8259493.51,	[July, August, September, October, No	2023
001-0000	GENERAL FUND	TRUE	[1781685.65, 1313486.58, 855540.05, 1	[1081674.54, 1535292.22, 1242577.51,	[5752010.17, 5530204.53, 5143167.07,	[July, August, September, October, No	2024
001-0000	GENERAL FUND	TRUE	[2413728.95, 1259751.38, 1040544.96,	[1180279.82, 1156989.73, 1067001.03,	[7378611.39, 7481373.04, 7454916.97,	[July, August, September, October, No	2025

Figure 2: Sample Cash Account History Report from USAS

- Solution: Developed a Python program that cleans and formats the data into an easy-to-read Excel file, automatically generates charts, and emails the completed report to the district Treasurer.
 - Uses pandas to isolate bracketed cells and extract their numeric values, then uses Openpyxl to write the cleaned data to a new Excel file and generate charts within that file.

Project 2 Bar Chart Example

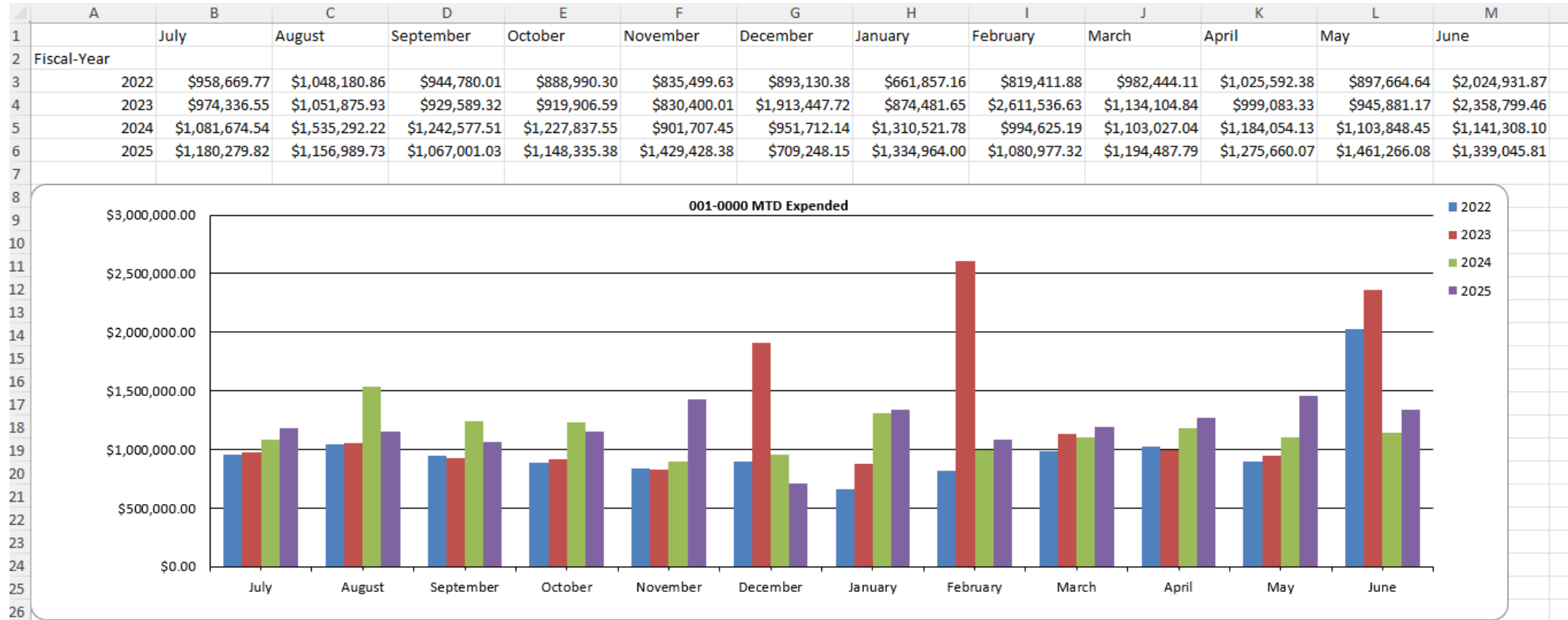


Figure 3: A bar chart of expenditures created after the script has finished running

Project 2 – Pivot Function

- Pandas has a unique and handy function called “Pivot” that does the following:
 - Reshapes Data: Rotates a long list into a horizontal grid.
 - Organizes Headers: Turns a column of repeating names into a single row of unique headers.
 - Aggregates Information: Automatically groups data (like monthly totals) into the correct cells, eliminating manual sorting.
- https://pandas.pydata.org/docs/user_guide/reshaping.html#reshaping-pivot

Project 2 – Pivot Function (cont.)

	A	B	C	D	E	F	G	H
1	Full Account	Description	Active	MTD Received	MTD Expected	Month End	Month	Fiscal Year
2	001-0000	GENERAL	TRUE	2054442	958669.8	8220181	July	2022
3	001-0000	GENERAL	TRUE	760288.4	1048181	7932289	August	2022
4	001-0000	GENERAL	TRUE	800024.1	944780	7787533	September	2022
5	001-0000	GENERAL	TRUE	838329.2	888990.3	7736872	October	2022
6	001-0000	GENERAL	TRUE	525231.9	835499.6	7426604	November	2022
7	001-0000	GENERAL	TRUE	518161.3	893130.4	7051635	December	2022
8	001-0000	GENERAL	TRUE	1395071	661857.2	7784848	January	2022
9	001-0000	GENERAL	TRUE	2094644	819411.9	9060081	February	2022
10	001-0000	GENERAL	TRUE	1289225	982444.1	9366862	March	2022
11	001-0000	GENERAL	TRUE	1201788	1025592	9543057	April	2022
12	001-0000	GENERAL	TRUE	487762.8	897664.6	9133155	May	2022
13	001-0000	GENERAL	TRUE	481224.1	2024932	7589447	June	2022
14	001-0000	GENERAL	TRUE	1995530	974336.6	8610641	July	2023
15	001-0000	GENERAL	TRUE	776006.5	1051876	8334772	August	2023
16	001-0000	GENERAL	TRUE	854310.9	929589.3	8259494	September	2023
17	001-0000	GENERAL	TRUE	1127263	919906.6	8466850	October	2023
18	001-0000	GENERAL	TRUE	571726.9	830400	8208177	November	2023
19	001-0000	GENERAL	TRUE	597249.6	1913448	6891979	December	2023
20	001-0000	GENERAL	TRUE	967400.4	874481.7	6984898	January	2023
21	001-0000	GENERAL	TRUE	1858519	2611537	6231880	February	2023
22	001-0000	GENERAL	TRUE	1700306	1134105	6798081	March	2023
23	001-0000	GENERAL	TRUE	1343465	999083.3	7142463	April	2023
24	001-0000	GENERAL	TRUE	620355.6	945881.2	6816937	May	2023
25	001-0000	GENERAL	TRUE	593861.5	2358799	5051999	June	2023
26	001-0000	GENERAL	TRUE	1781686	1081675	5752010	July	2024
27	001-0000	GENERAL	TRUE	1313487	1535292	5530205	August	2024
28	001-0000	GENERAL	TRUE	855540.1	1242578	5143167	September	2024
29	001-0000	GENERAL	TRUE	1232545	1227838	5147874	October	2024
30	001-0000	GENERAL	TRUE	667375.5	901707.5	4913542	November	2024
31	001-0000	GENERAL	TRUE	652681.9	951712.1	4614512	December	2024

Before

```
pivot_df = input_df.pivot(index="Fiscal-Year", columns="Month", values = "MTD Expended")
```

After

[illegible]

Project 2 – Example Line Graph Code

```
#Example of how to create a line chart
def example_line_chart(output_file:str):
    print("Example Chart")
    # Sample data
    data = [
        ["Month", "2023"],
        ["Jan", 10],
        ["Feb", 15],
        ["Mar", 12],
        ["Apr", 18],
        ["May", 20],
    ]

    wb = Workbook()
    ws = wb.active

    for row in data:
        ws.append(row)

    data_values = Reference(ws, min_col=2, min_row=1, max_col=2, max_row=len(data))
    categories = Reference(ws, min_col=1, min_row=2, max_col=1, max_row=len(data))

    chart = LineChart()
    chart.title = "Example Line Chart"
    chart.style = 20
    chart.add_data(data_values)
    chart.set_categories(categories)

    ws.add_chart(chart, "D1") # Place chart in Cell D1
    wb.save(output_file)
    print("example.xlsx created. Open it to see the chart.")
```

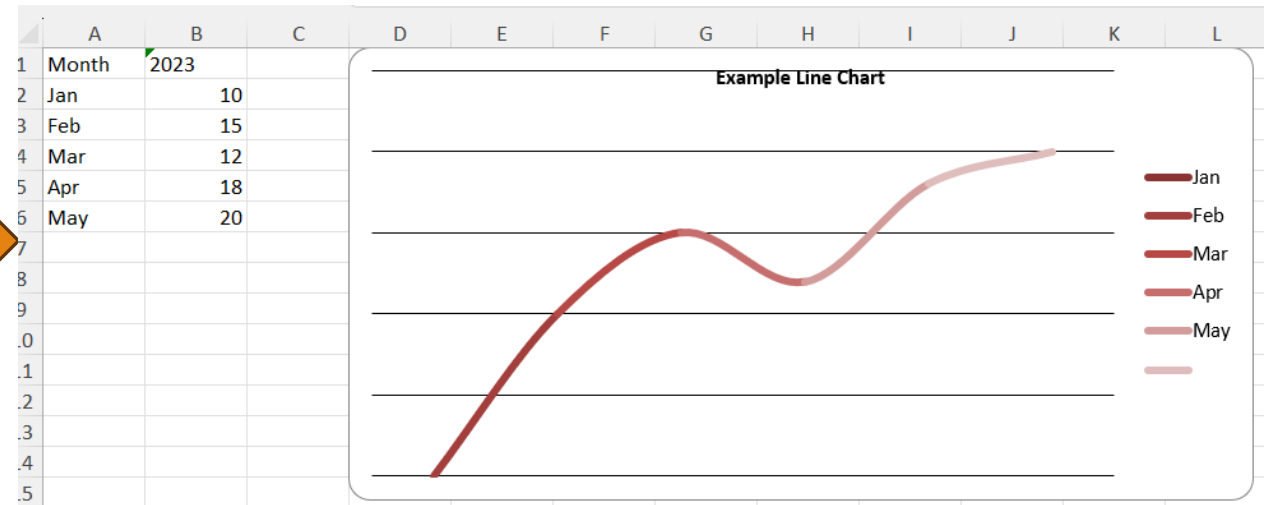
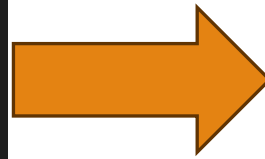


Figure 5: Output of example code

Figure 4: Example code on how to create a line chart using Python and Openpyxl charts library

Automated Workflow

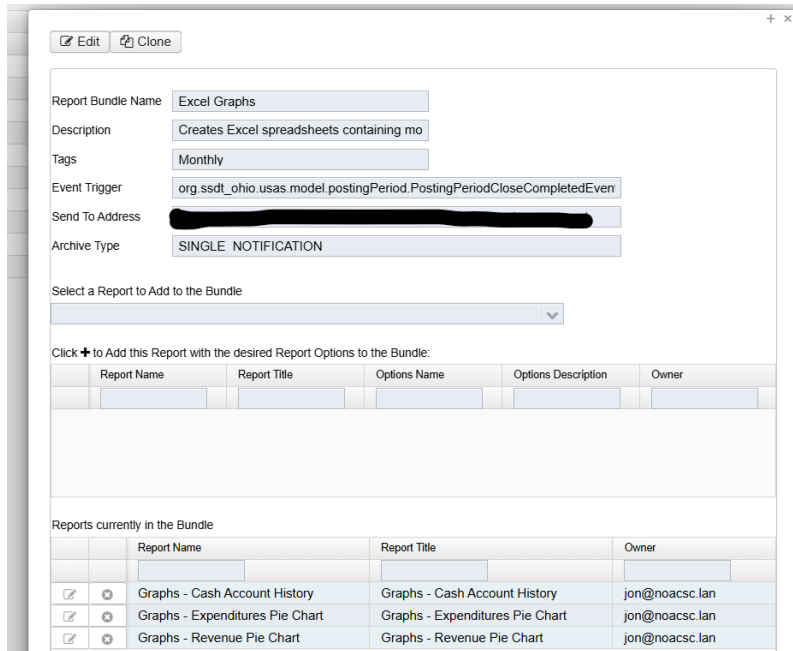


Figure 6: Screenshot of Report Bundle used for graphs

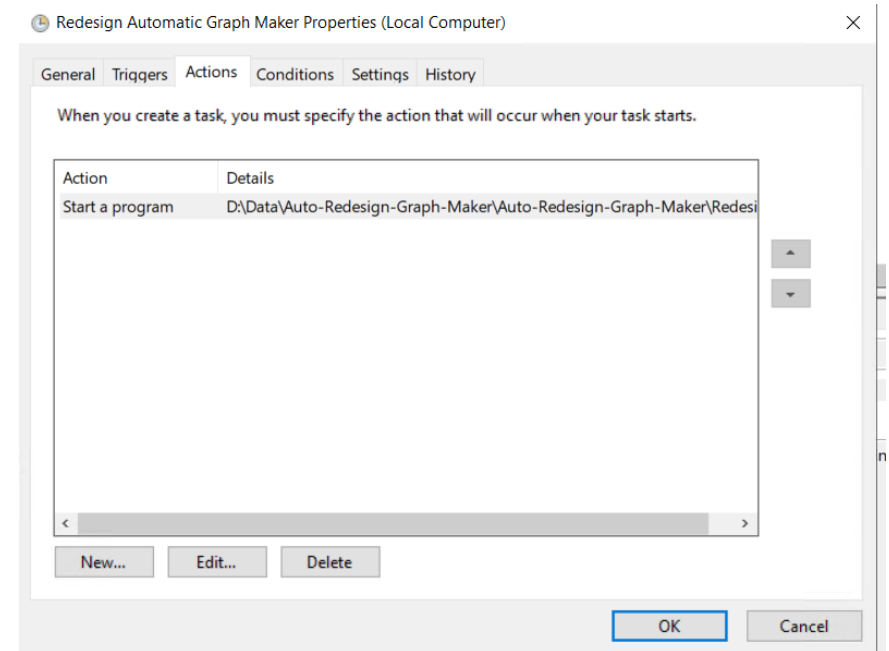


Figure 7: Screenshot of Windows Task Scheduler

- When a treasurer closes a month, a report bundle runs sending files to one of our servers via FTP.
- Windows Task Scheduler is used to set when the Python Scripts should run.

Using AI to Create Python Scripts

- AI tools can generate Python scripts from plain-language instructions, allowing anyone to automate tasks without deep programming knowledge.
- AI can suggest libraries, write functions, debug code, and optimize scripts based on best practices.
 - Some libraries documentation are hundreds of pages long. AI can break down just the functions you need.
- AI tools for Programming:
 - - Copilot
 - - Google Gemini
 - - ChatGPT

Using AI to Create Python Scripts (cont.)

- Provide clear and precise prompts of what you want the script to do.
- List the layout of your input and output files. If there are specific headers you want the script to look through list them in the prompt.
- Ask it to give step by step instructions and to include comments explaining the code
- DO NOT ENTER SENSITIVE INFORMATION IN THE PROMPT!
- Remember, AI is not always accurate. You will probably have to make slight changes to the AI provided code to fit your needs.

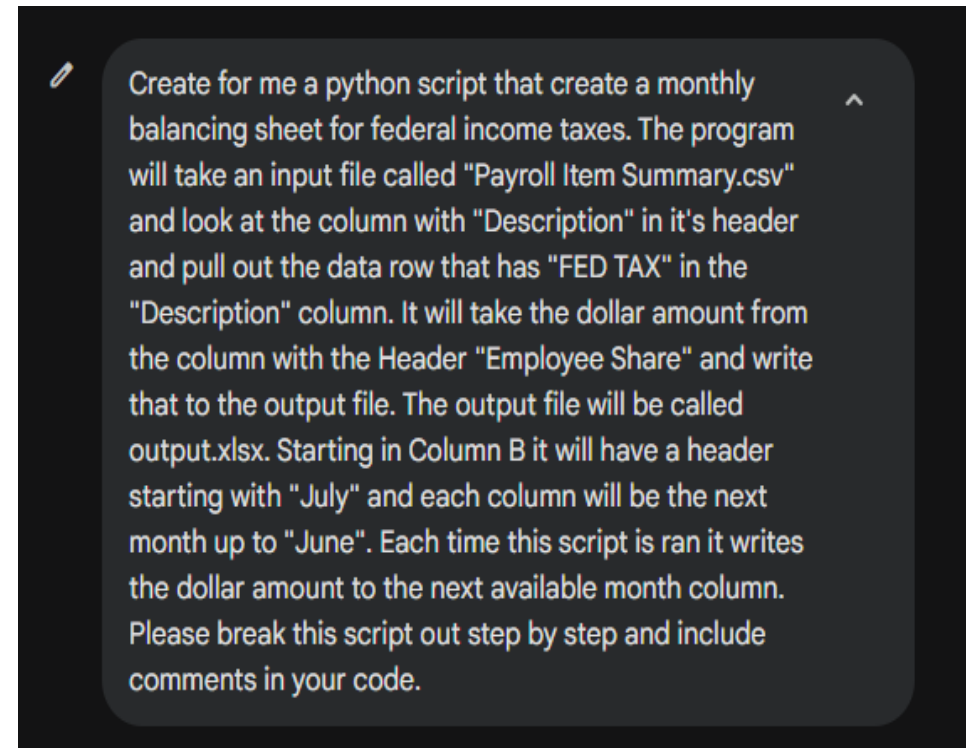


Figure 8: Example of entering a prompt in Gemini

Git Repo

Full projects are available on Github: <https://github.com/JPyles01/OECN-2026-Projects-Repository/tree/main/OECN-2026-Automating-Excel-Spreadsheets-Public>



If you have any questions, you can contact me at: ion@noacsc.org